Khwezi Mining Monitoring Application

Flowcharts for All Functions

MINN2020A: Computer Programming for Mining

19 October 2026

Key: rounded box = start/end; parallelogram = input/output; rectangle = process; diamond = decision; red box = error message and exit.

1 Engineering rules

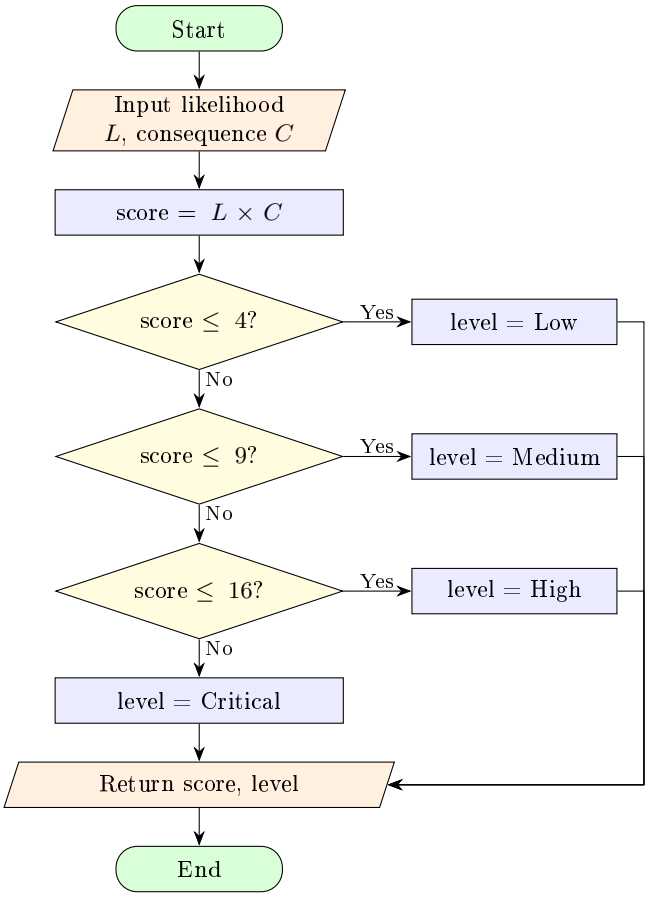


Figure 1: *risk_classification*: risk score and level.

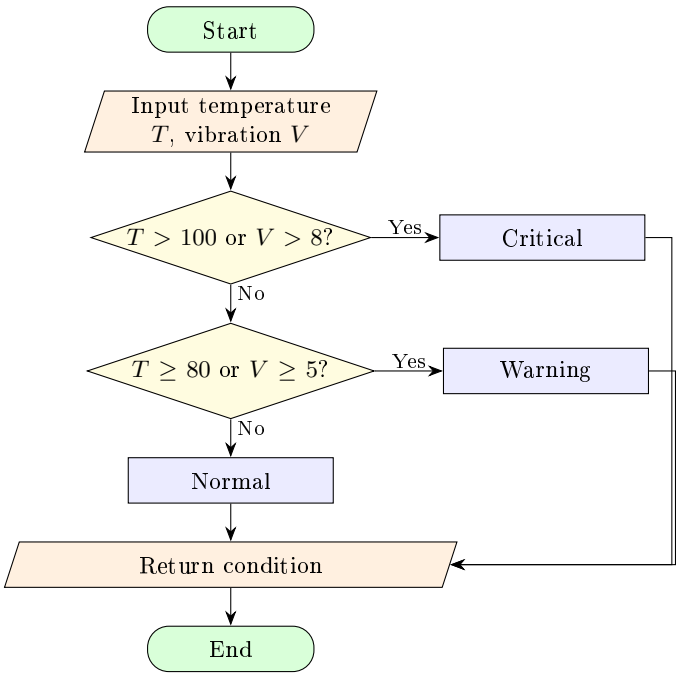


Figure 2: *equipment_ condition*: prototype thresholds.

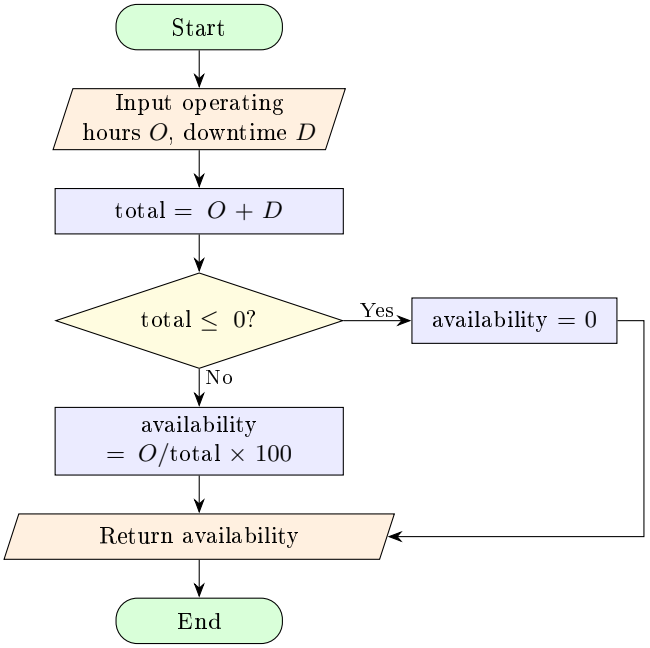


Figure 3: *availability_ percent*: equipment availability.

2 Access control

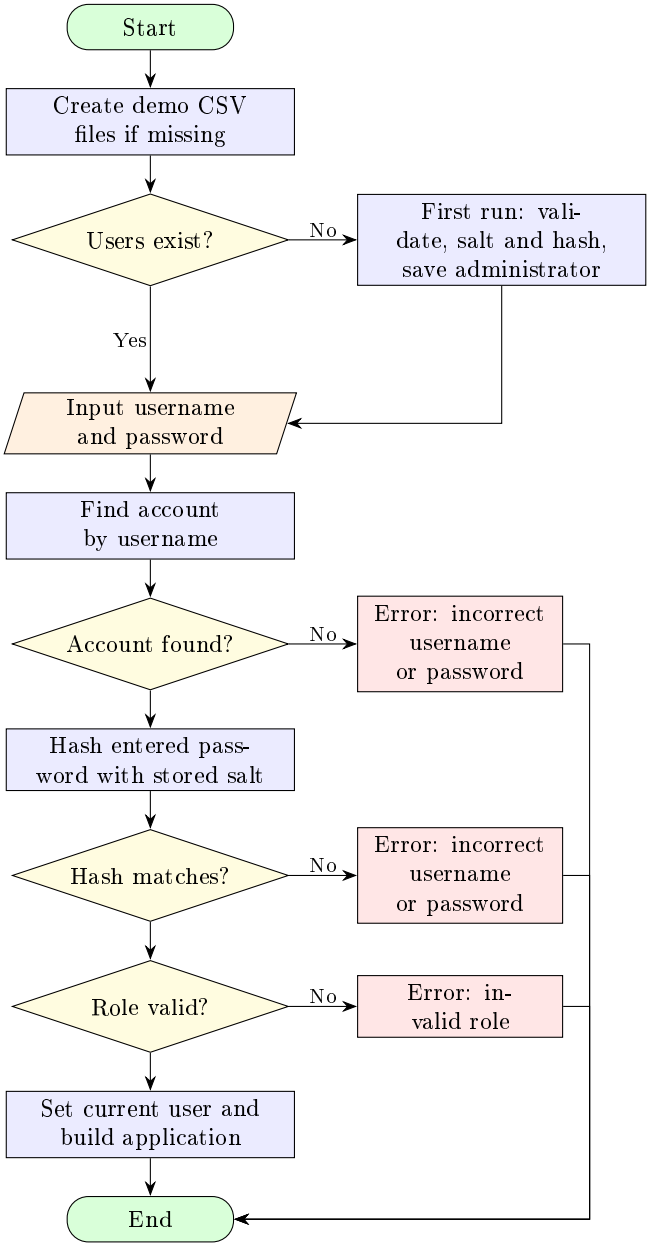


Figure 4: *create_first_admin* and *login*.

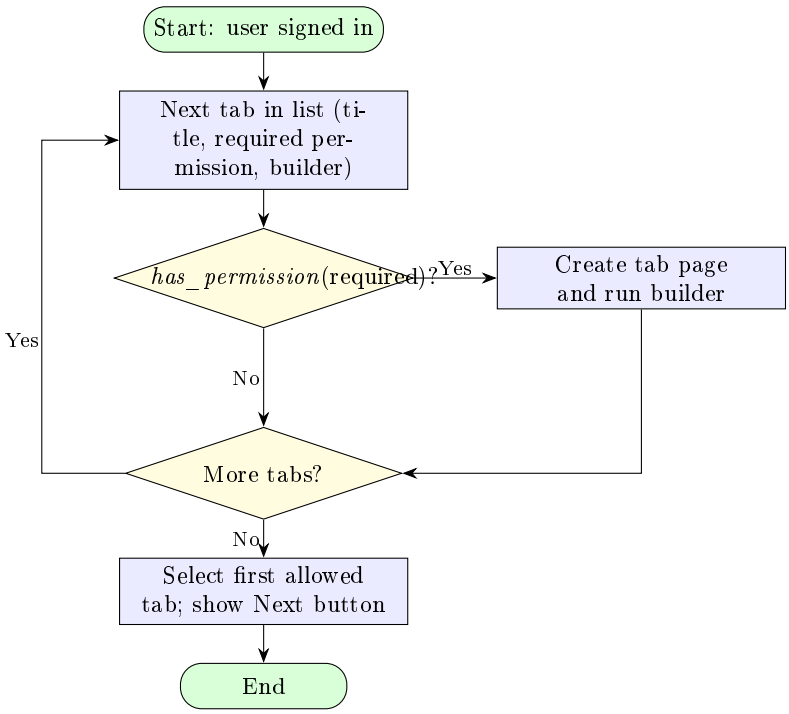


Figure 5: *has_permission* and role-filtered navigation (*build_app*).

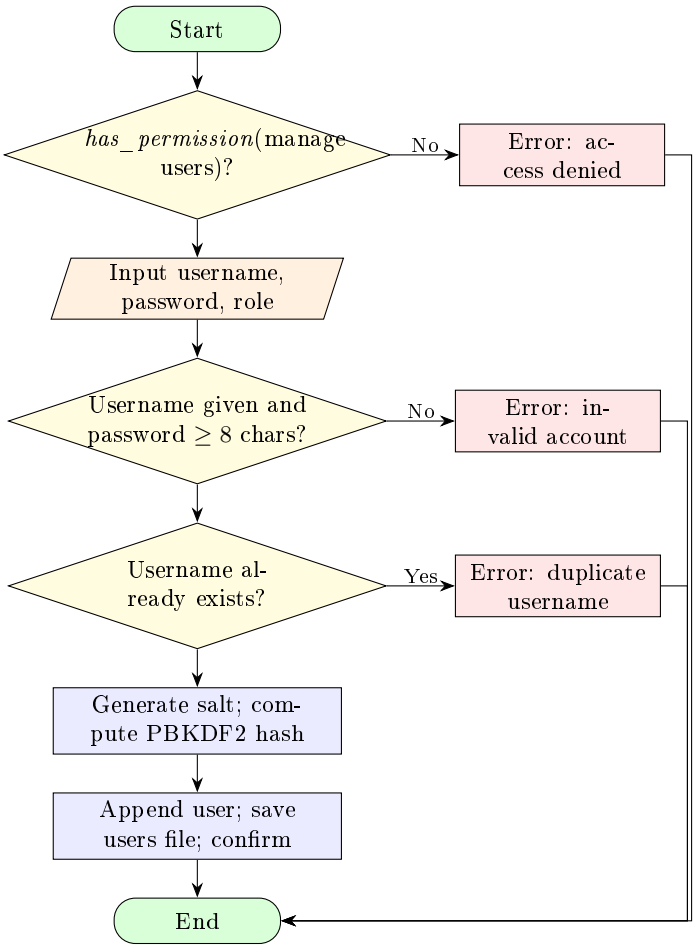


Figure 6: *add_user*: administrator creates an account.

3 Alerts and dashboard

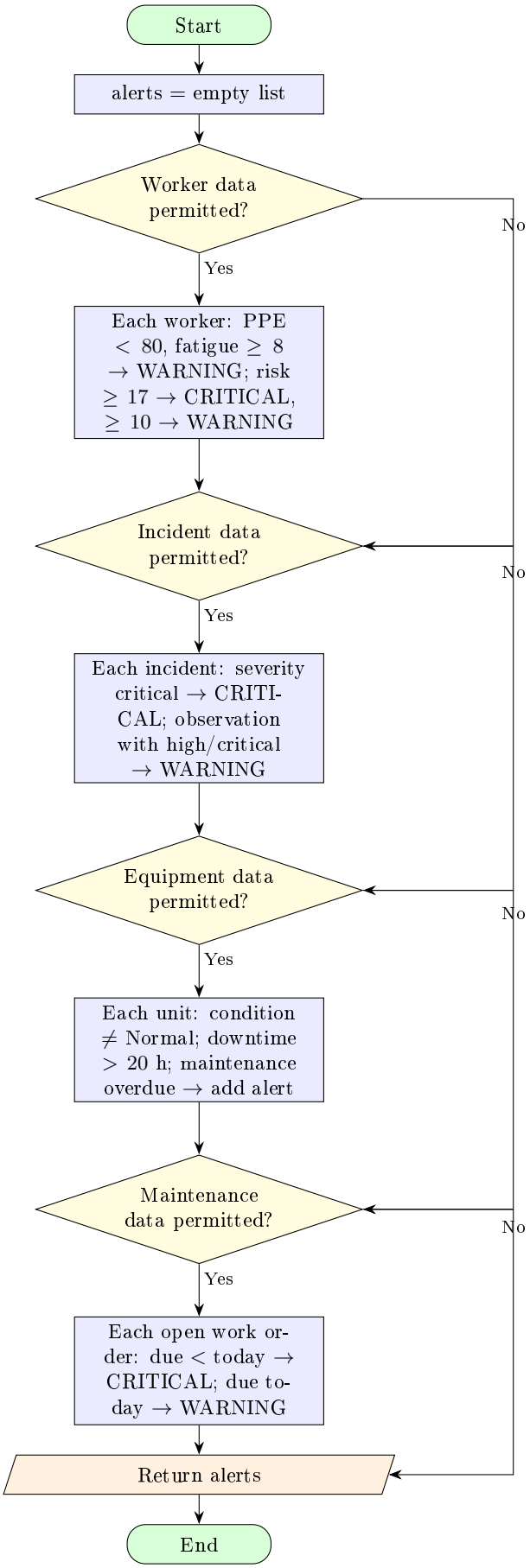


Figure 7: *collect_ alerts*: automatic alert generation.

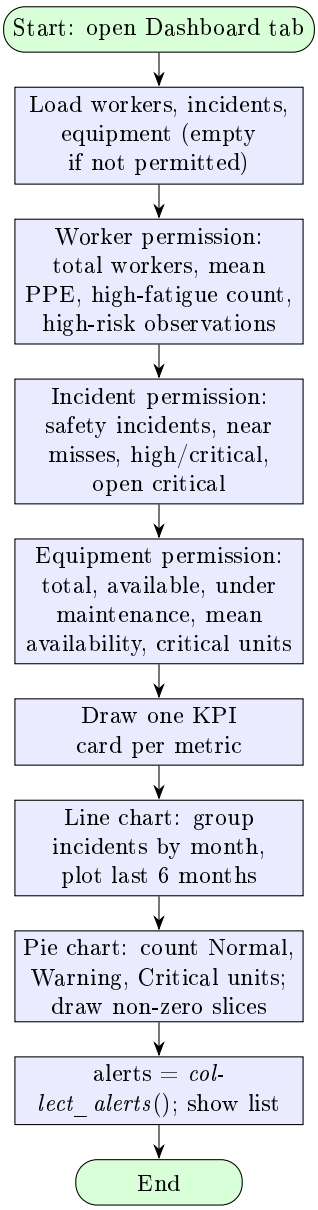


Figure 8: *build_dashboard*: KPI cards, charts and alerts.

4 Worker safety and incidents

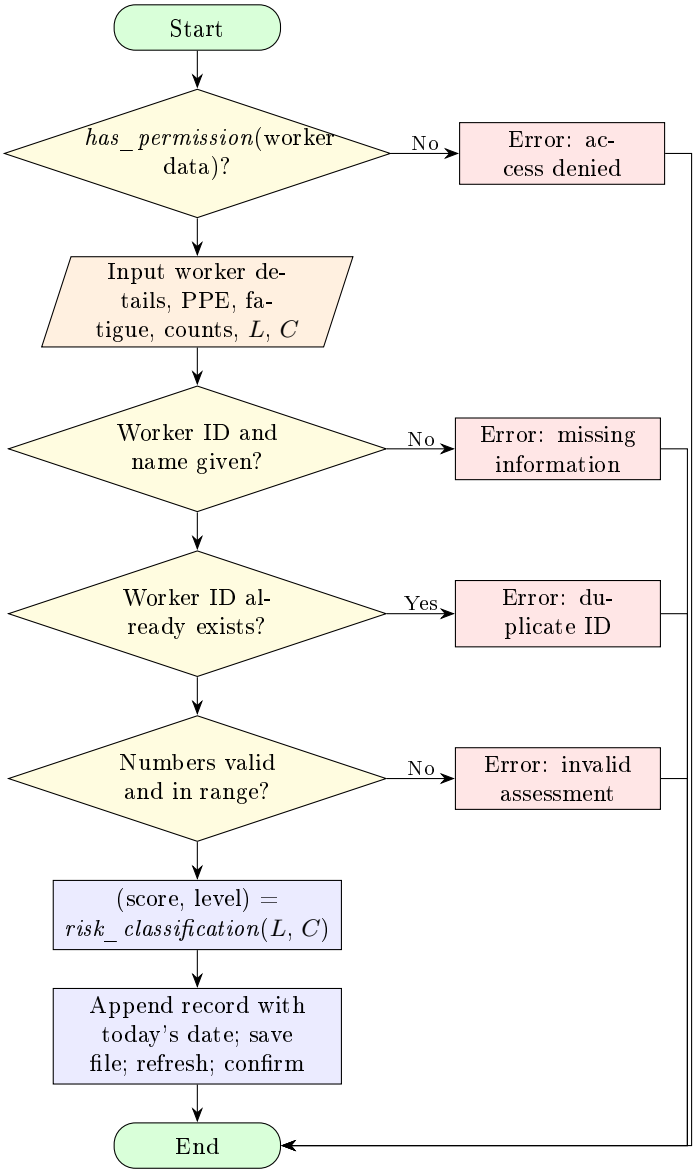


Figure 9: *add_worker_assessment.*

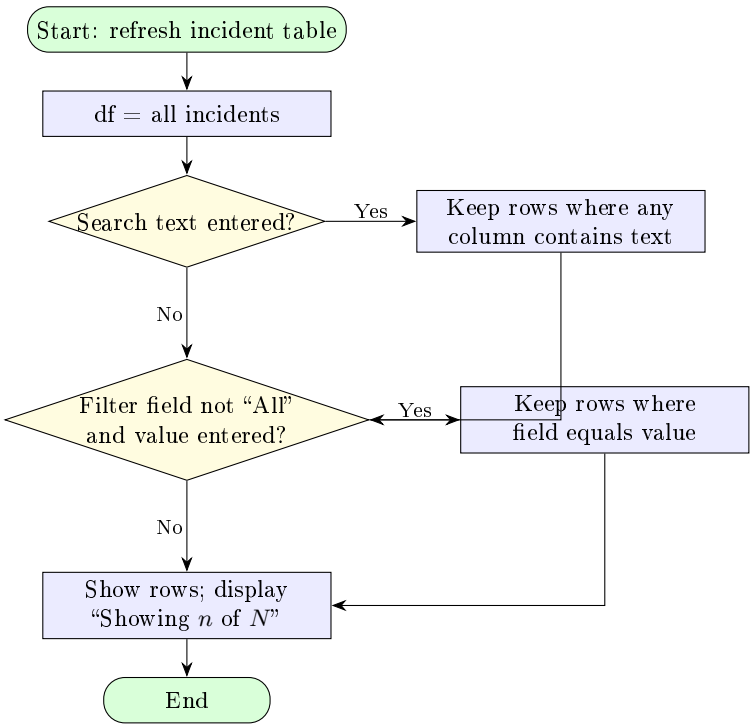


Figure 10: *refresh_ incident_ table*: search and filter.

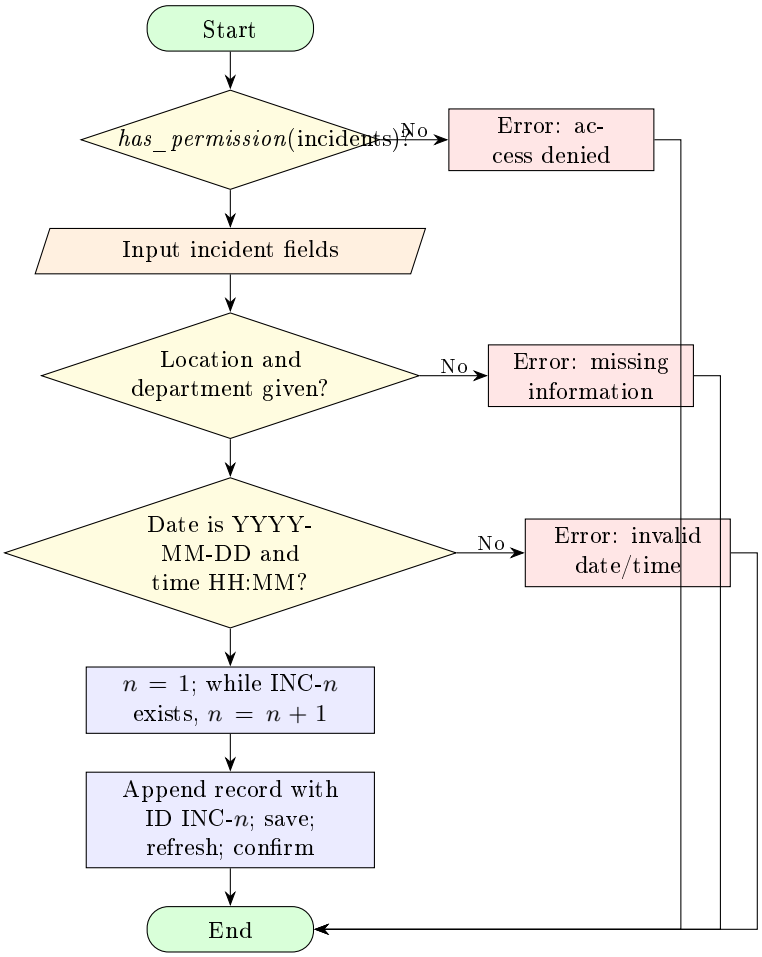


Figure 11: *add_ incident*.

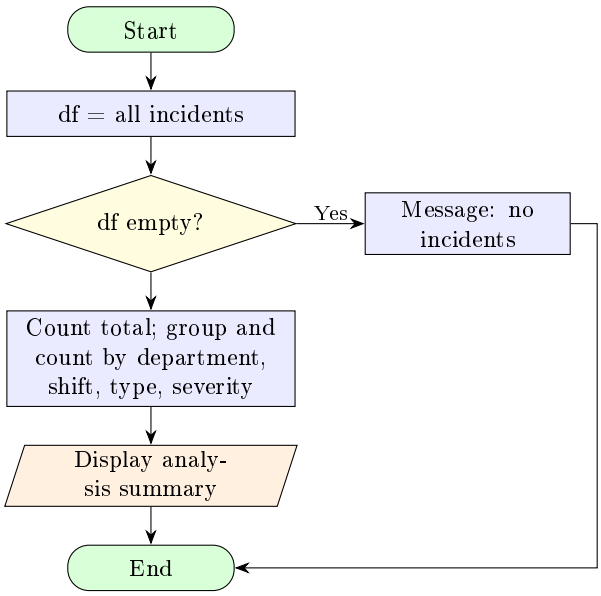


Figure 12: *incident_ analysis.*

5 Equipment and maintenance

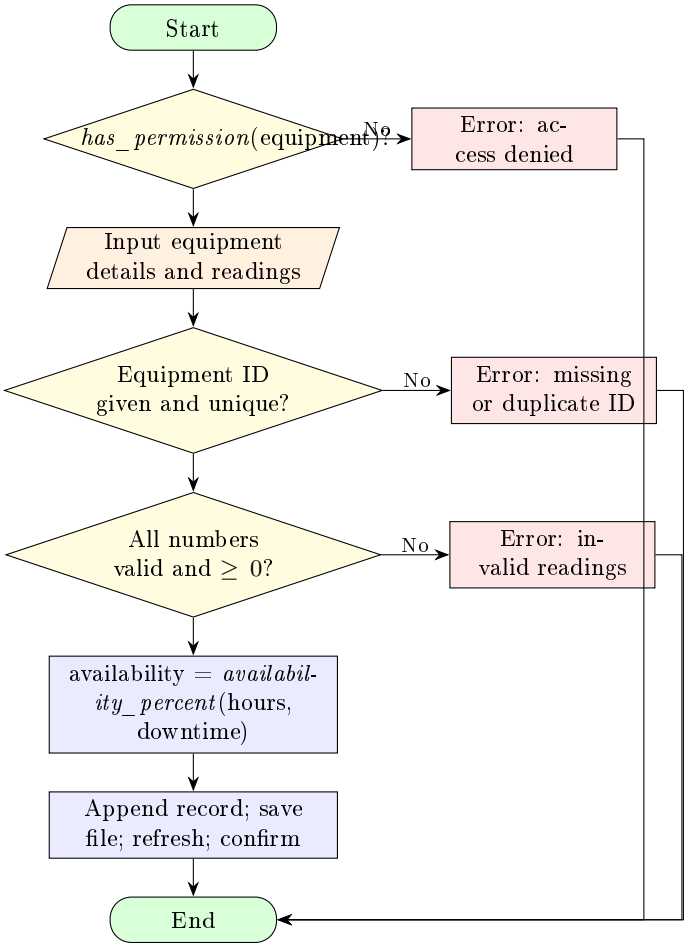


Figure 13: *add_equipment*.

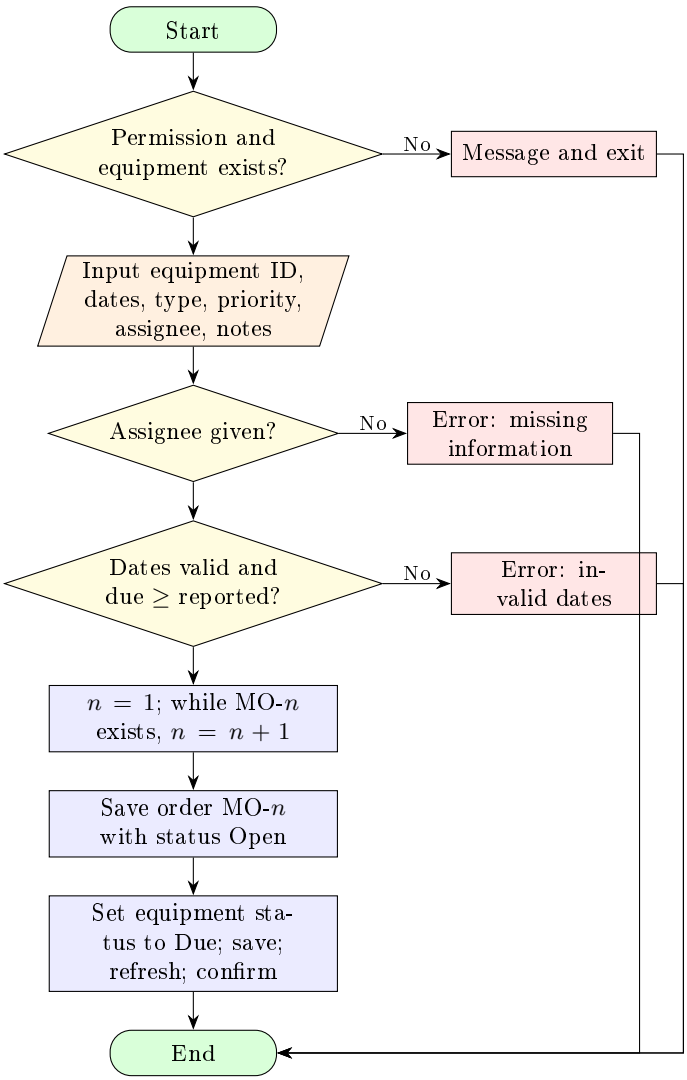


Figure 14: *add_ maintenance_ order.*

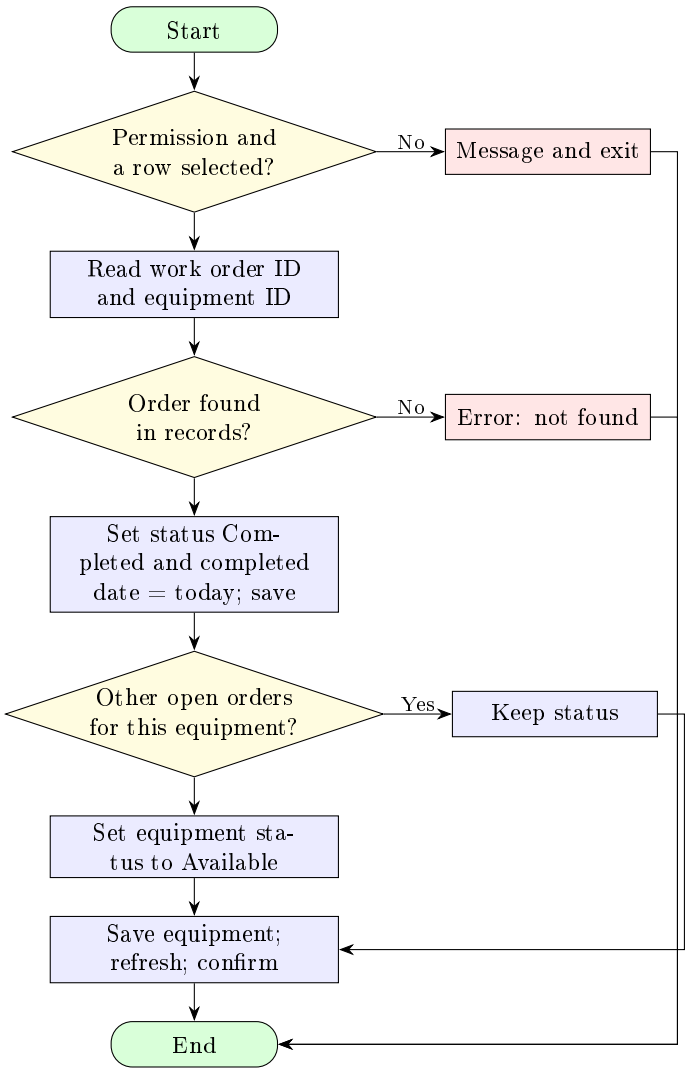


Figure 15: *complete_ maintenance_ order.*

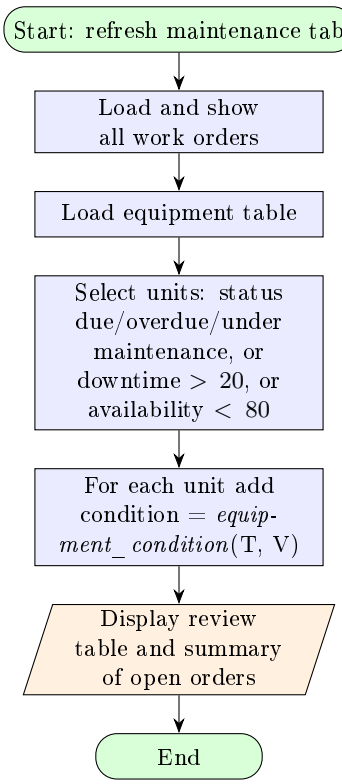


Figure 16: *refresh_maintenance_table*: equipment needing review.

6 Risk assessment, reports and external data

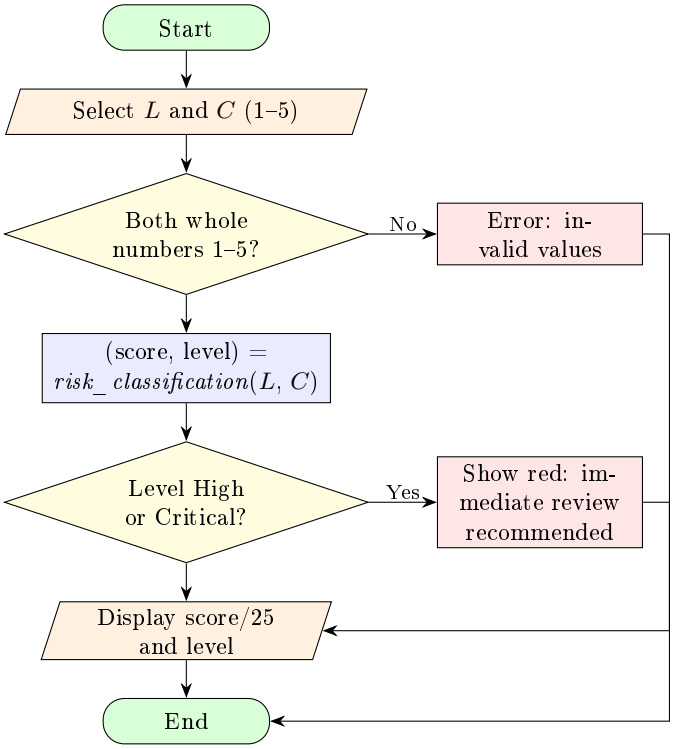


Figure 17: Risk calculator (*build_risk*).

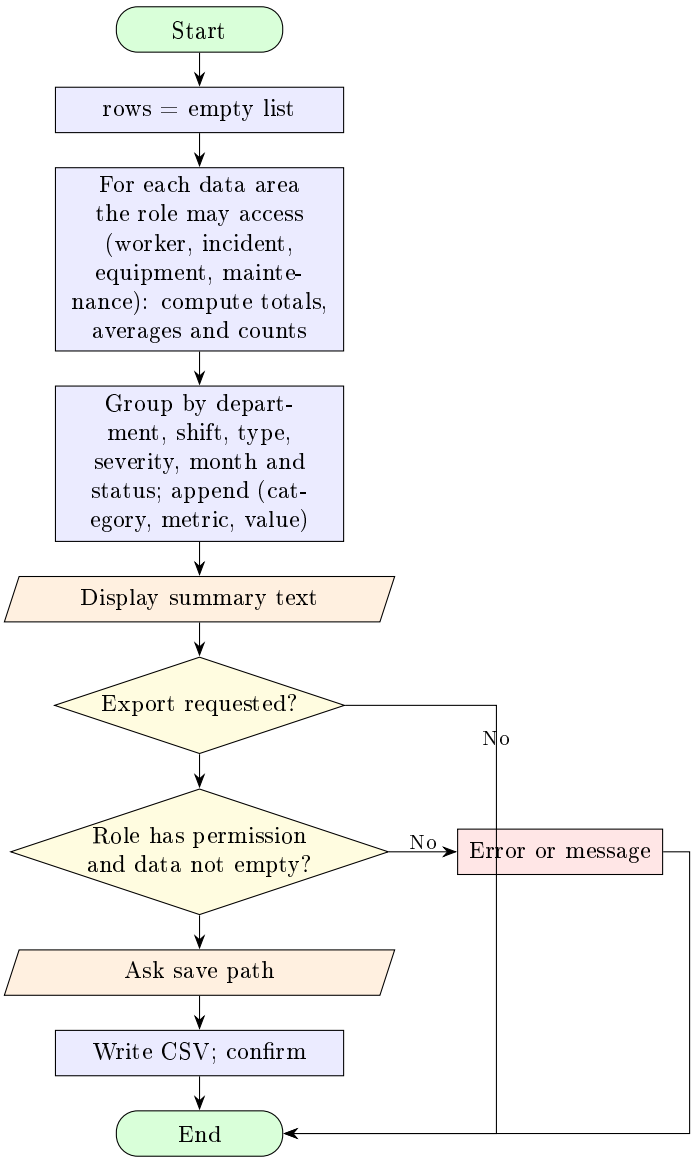


Figure 18: *report_rows*, *generate_summary*, *export_dataset*.

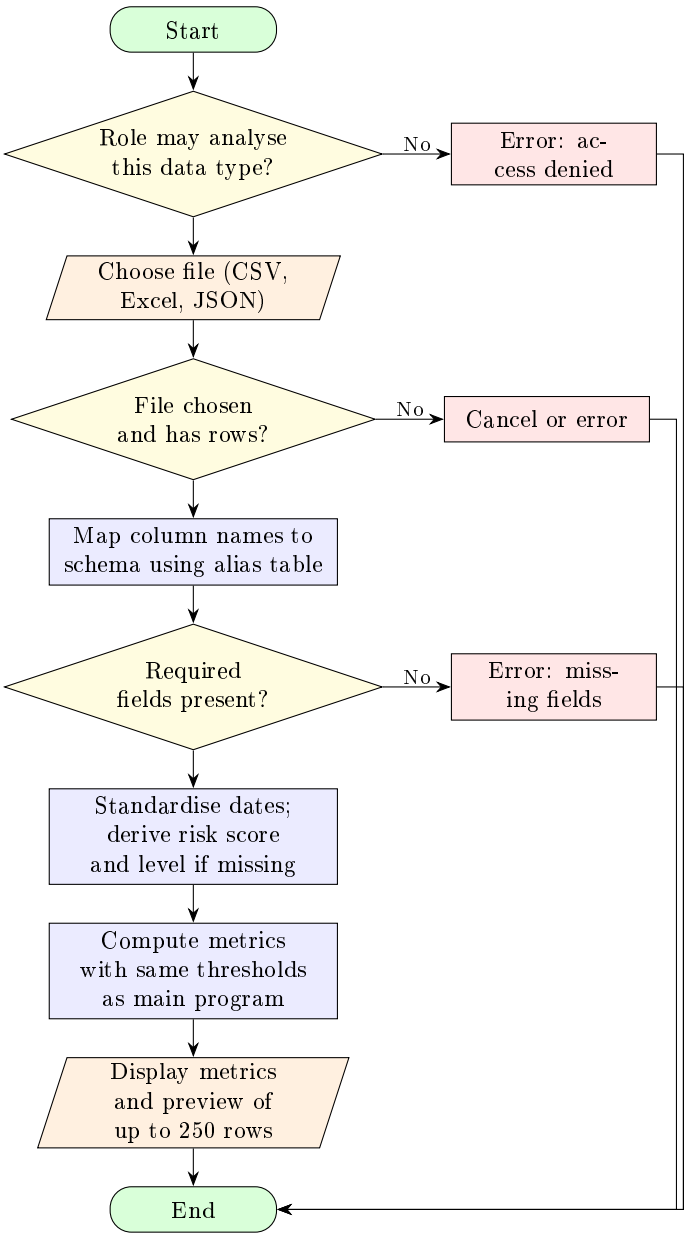


Figure 19: *upload_external_data*: analysis of uploaded files (saved records unchanged).